



CALCULATOR USING C LANGUAGE

A Personal Project Developed Using C
Language

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ABSTRACT

This project presents a console-based calculator developed using the C programming language. The application is designed to perform basic mathematical operations, including addition, subtraction, multiplication, division, power, modulus, and square calculation. It uses a menu-driven interface that allows users to select an operation, enter the required inputs, and view the result. The program also includes error handling for invalid operations and division or modulus by zero, making it more reliable and user-friendly.

The main objective of this project is to strengthen the understanding of fundamental C programming concepts such as variables, operators, loops, conditional statements, switch-case, user input handling, and the use of the **math.h** library. This project demonstrates logical problem-solving and serves as a practical application of core programming skills.

INTRODUCTION

A calculator is one of the most commonly used applications for performing mathematical calculations quickly and accurately. This project was developed using the C programming language to demonstrate the implementation of basic arithmetic operations in a console-based environment.

The calculator provides a simple menu-driven interface that allows users to perform operations such as addition, subtraction, multiplication, division, power, modulus, and square calculation. The program also includes error handling for invalid operations and division or modulus by zero, ensuring reliable execution.

This project helped improve my understanding of fundamental C programming concepts, including variables, data types, operators, conditional statements, switch-case, loops, user input handling, and the use of the **math.h** library. It also enhanced my logical thinking and problem-solving skills while developing a practical application.

OBJECTIVE

The primary objective of this project is to develop a console-based calculator using the C programming language and to apply fundamental programming concepts in a practical application. The project aims to perform basic arithmetic operations such as addition, subtraction, multiplication, division, power, modulus, and square calculation through a simple menu-driven interface.

This project also focuses on improving problem-solving skills and gaining hands-on experience with C programming concepts, including variables, operators, conditional statements, switch-case, loops, user input handling, and the use of the **math.h** library. Additionally, it helps in understanding how to write structured, efficient, and user-friendly programs.

FEATURES

The calculator application provides the following features:

- Performs addition, subtraction, multiplication, and division.
- Calculates the power of a number using the **math.h** library.
- Supports modulus operation for integer values.
- Calculates the square of a number.
- Provides a menu-driven interface for easy user interaction.
- Allows users to perform multiple calculations using a do-while loop.
- Displays appropriate error messages for invalid operations.
- Prevents division by zero and modulus by zero through error handling.

TOOLS AND TECHNOLOGIES USED

The following tools and technologies were used to develop this project:

- Programming Language: C
- Development Environment (IDE): Dev-C++
- Compiler: GCC Compiler
- Header Files: **stdio.h**, **math.h**
- Platform: Windows

ALGORITHM

1. Start the program.
2. Display the calculator menu with available operations.
3. Declare variables for numbers, operator, result, and user choice.
4. Repeat the following until the user chooses to exit:
 - Read the first number.
 - Read the operator.
 - Read the second number (if required).
 - Use a switch statement to perform the selected operation.
 - If the operation is division or modulus, check whether the second number is zero.
 - Display the result or an error message.
 - Ask the user whether they want to continue.
5. If the user enters Y or y, repeat the process.
6. Otherwise, display a thank-you message.
7. End the program.

SOURCE CODE

```
//To run actual calculator
#include<stdio.h>
#include<math.h>

int main ()
{
    printf("\nCALCULATOR MENU\n");
    printf("Addition: +\n");
    printf("Subtraction: -\n");
    printf("Multiplication: *\n");
    printf("Division: /\n");
    printf("Power: ^\n");
    printf("Modulus: %%\n");
    printf("Square: s\n");
    float a;
    char operation;
    float b;
    float c;
    char ch;
    do
    {
        printf("\nEnter expression\n");
        scanf("%f",&a);
        scanf(" %c",&operation);
        scanf("%f",&b);
        switch(operation)
        {
            case '+':
                c=a+b;
                printf("%f",c);
                break;
            case '-':
                c=a-b;
                printf("%f",c);
                break;
```

```

case '*':
    c=a*b;
    printf("%f",c);
    break;
case '/':
    if(b==0)
    {
        printf("Error!Division by Zero is not allowed");
    }
    else
    {
        c=a/b;
        printf("%f",c);
    }
    break;
case '^':
    c=pow(a,b);
    printf("%f",c);
    break;
case '%':
    if((int)b == 0)
    {
        printf("Error!Modulus by zero is not allowed.");
    }
    else
    {
        printf("%d", (int)a % (int)b);
    }
    break;
case 's':
    b = a;
    c=a*b;
    printf("%f",c);
    break;
default:
    printf("Error!Such operation is not allowed.");

```

```
                break;
            }
            printf("\nDo you want to continue? (Y/N) :");
            scanf(" %c", &ch);
        }
        while(ch=='Y' || ch=='y');
        printf("\nThank you for using Calculator!");
        return 0;
    }
}
```


OUTPUT SCREENSHOTS

Figure 1: Calculator Menu

```
CALCULATOR MENU
Addition: +
Subtraction: -
Multiplication: *
Division: /
Power: ^
Modulus: %
Square: s
```

Figure 2: Addition Operation

```
Enter expression
2+5
7.000000
Do you want to continue? (Y/N) :y
```

Figure 3: Subtraction Operation

```
Enter expression
2-5
-3.000000
Do you want to continue? (Y/N) :y
```

Figure 4: Multiplication Operation

```
Enter expression
2*5
10.000000
Do you want to continue? (Y/N) :y
```

Figure 5: Division Operation

```
Enter expression
2/5
0.400000
Do you want to continue? (Y/N) :y

Enter expression
2/0
Error!Division by Zero is not allowed
Do you want to continue? (Y/N) :y
```

Figure 6: Power Operation

```
Enter expression
2^5
32.000000
Do you want to continue? (Y/N) :y
```

Figure 7: Modulus Operation

```
Enter expression
2%5
2
Do you want to continue? (Y/N) :y

Enter expression
2%0
Error!Modulus by zero is not allowed.
Do you want to continue? (Y/N) :y
```

Figure 8: Square Operation

```
Enter expression
2s2
4.000000
Do you want to continue? (Y/N) :n

Thank you for using Calculator!
```

CONCLUSION

The Calculator Using C Language project successfully demonstrates the implementation of fundamental programming concepts through a practical application. The program performs basic arithmetic operations such as addition, subtraction, multiplication, division, power, modulus, and square calculation using a simple menu-driven interface.

During the development of this project, I gained practical experience in using variables, operators, switch-case statements, loops, user input handling, and the math.h library. Error handling for division by zero and invalid operations was also implemented to improve the reliability of the application.

Overall, this project strengthened my understanding of C programming and enhanced my logical thinking and problem-solving skills. It also serves as a strong foundation for developing more advanced applications in the future.

FUTURE ENHANCEMENTS

The calculator project can be enhanced further by implementing additional features and improving the overall user experience. Some possible future enhancements include:

- Develop a graphical user interface (GUI) for a better user experience.
- Implement user-defined functions to improve code modularity and readability.
- Add scientific calculator functions such as square root, logarithm, sine, cosine, and tangent.
- Maintain a history of previous calculations.
- Improve input validation and error handling for invalid user inputs.
- Extend the application to support advanced mathematical operations.